seqCNA Output

The results of a seqCNA run are delivered in the folder `seqcna/`. `Proj_<no>` below is the project number. Each sample is named by its tumor sample ID (`s_...`). The normal that each tumor was compared with is chosen by the pipeline (Methods, "Choice of the normal") and is not named in the files.

File	Content
`Proj_<no>___seqSeg.pdf`	Copy number plot of each tumor, one per page
`Proj_<no>___IGV.seg`	Every segment of every tumor, for IGV
`Proj_<no>___GeneTable.xlsx`	Called genes, one row per tumor and gene
`GeneMatrix.csv`	Called genes, a matrix of genes by tumors
`SegmentMatrix.csv`	Called genes with the segment they are in
`output.pdf`, `output.html`	This document
`methods.pdf`, `methods.html`	The methods

1 Log2 ratios

Every copy number value in the files is a log2 ratio of tumor to normal coverage, after GC correction and scaling for the number of reads, shifted so that one level of the tumor, the baseline, is 0 (Methods, "Segment clustering and the baseline"). A positive value is a gain relative to the baseline, a negative value a loss.

seqCNA does not estimate the purity (the fraction of tumor cells in the sample) or the ploidy of the tumor. So the log2 ratios are not converted to numbers of copies, and the baseline is not known to be two copies.

A gene is called when its segment has FDR < 0.05 and an absolute log2 ratio greater than 1, that is a coverage ratio more than twice or less than half that of the baseline (Methods, "Gene-level calls"). The bound is strict: a segment at exactly 1 or -1 is not called. The threshold is on the log2 ratio only. Smaller changes are not called, even with FDR < 0.05; they can be seen in the plots and the IGV file.

2 `Proj_<no>___seqSeg.pdf`

One page per tumor, titled with the tumor ID.

- The x axis is the position along the genome in Mb, with the chromosomes end to end. Dashed vertical lines separate the chromosomes, and their numbers are at the bottom. Chromosome X is numbered 23 (human) or 20 (mouse).
- Each point is the log2 ratio of one bin, light blue and gray on alternating chromosomes. The y axis runs from -5 to 5; points beyond that are not shown. In targeted sequencing there are bins only in the capture targets, so the points are not spread evenly along the genome.
- The red lines are the segments, at the level of their cluster.
- The dashed horizontal lines are at log2 ratios -1, log2(1.5) = 0.58 and 1. Only segments below -1 or above 1 can be called.
- The dotted horizontal lines are at plus and minus the MAD.

The two lines of text at the top are measures of the data of the pair:

Value	Meaning
`NumBins`	Number of bins segmented
`MAD`	Median absolute deviation (R `mad`, scaled to estimate a standard deviation) of the bin log2 ratios from their segment level: the noise in the data
`nSegs`	Number of segments, chromosome X included
`RMSD.Segs.noX`	Root mean square of the segment levels, chromosome X excluded; each segment counts once, whatever its length
`sum.logr.sq`	Sum of the squared bin log2 ratios
`rms.logr`	Root mean square of the bin log2 ratios. The normal with the lowest value is the one chosen
`rms.logr.flat`	Root mean square of the bin log2 ratios around their segment level, another measure of the noise

Chromosome X is plotted but never called. If the tumor and the normal come from individuals of different sex, the whole of X is shifted down (male tumor, female normal) or up (female tumor, male normal).

3 `Proj_<no>___IGV.seg`

Every segment of every tumor, significant or not, tab-separated, one row per segment. To view it in IGV, choose the genome of the project (hg19 for GRCh37/b37, mm10 for mouse), then File > Load from File. IGV colors each segment by its log2 ratio, red for a gain and blue for a loss.

Column	Meaning
`ID`	Tumor ID
`chrom`	Chromosome: 1 to 22 and X (mouse: 1 to 19 and X), without `chr`
`loc.start`, `loc.end`	Centers of the first and last bins of the segment, in bp
`num.mark`	Number of bins in the segment
`seg.mean`	Log2 ratio of the segment (the level of its cluster)

The p-value and FDR of each segment are not in this file; for the segments with a called gene they are in `SegmentMatrix.csv` and the gene table.

4 Gene call tables

The three tables list the genes called in at least one tumor (Methods, "Gene-level calls"), on the autosomes only. In each tumor, a gene is given one segment: of the segments that its exons overlap, the one with the lowest log2 ratio, significant or not. The gene is called if that segment has FDR $<$ 0.05 and an absolute log2 ratio greater than 1. So a gene partly in a deletion is called as a deletion, and a gene partly in a gain is not called if the rest of it is in a segment with a lower log2 ratio. If no gene is called in any tumor, the tables are not written.

Genes are named by their gene symbol and the Ensembl ID of the canonical transcript used.

4.1 `Proj_<no>___GeneTable.xlsx`

One sheet, one row per tumor and called gene, sorted by tumor and position.

Column	Meaning
`ID`	Tumor ID
`gene`, `transcript`	Gene symbol and Ensembl transcript
`chrom`, `loc.start`, `loc.end`	The segment, as in the IGV file
`seg.mean`	Log2 ratio of the segment, to 3 decimals
`FDR`	FDR of the segment

4.2 `GeneMatrix.csv`

One row per called gene. The columns are `gene`, `transcript`, then one column per tumor with at least one call. A value is the log2 ratio of the call, or 0 if the gene is not called in that tumor. A 0 means "not called", not a log2 ratio of 0: the gene can still be in a segment with a log2 ratio away from 0 that does not pass the threshold. Tumors without any call have no column.

4.3 `SegmentMatrix.csv`

The same calls with their segments, one row per called gene and segment, sorted by position. A row is the call of one tumor (or of several tumors with the same segment); its column holds the log2 ratio, and the columns of the other tumors are empty.

Column	Meaning
`chrom`, `loc.start`, `loc.end`, `num.mark`	The segment, as in the IGV file
`cluster`	Number of the segment's cluster within the tumor; for checking only

Column	Meaning
`pValue`, `FDR`	p-value and FDR of the segment
`start`, `stop`, `exon`	An exon of the gene that overlaps the segment
`gene`, `transcript`	Gene symbol and Ensembl transcript
one column per tumor	Log2 ratio of the segment, in the rows of that tumor's calls